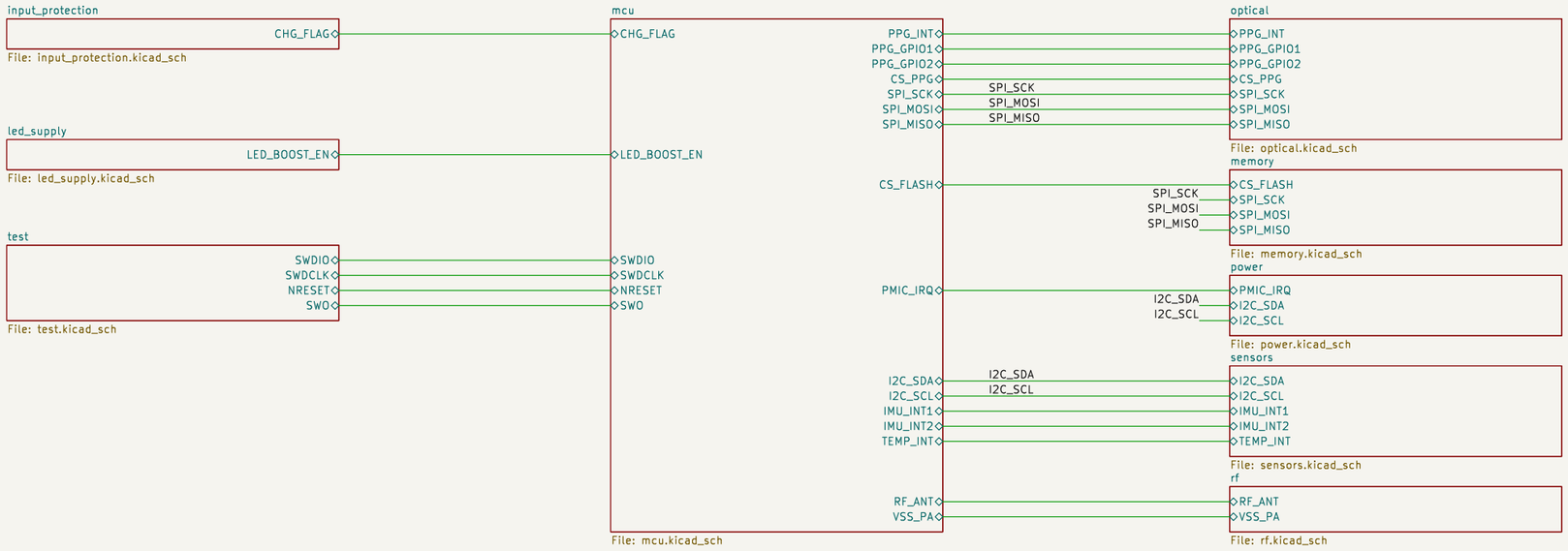
HEALTHOS BAND REV 1.1 — main board, top level

Signals flow left to right between sheets. Supply rails (VBUS_PROT, VSYS, 1.8 V, VLED, ground) are power symbols inside the sheets and need no wire here.



How to read this page: each box is one sheet. A wire between two boxes is a signal that crosses sheets. SPI_SCK, SPI_MOSI and SPI_MISO serve the optical sheet and the memory sheet; I2C_SDA and I2C_SCL serve the sensors sheet and the power sheet. The second receiver of each of these five nets is joined by name (label) because a wire would cross other signals.

Power: the charger input (VBUS_PROT) from the charging input sheet, the battery, VSYS and the 1.8 V rail are power symbols, so they connect by name across sheets and do not appear as wires here. The nPM1300 on the power sheet makes VSYS and the 1.8 V rail.

Done by Harsh Thakur

Sheet: /
File: smartwatch.kicad_sch

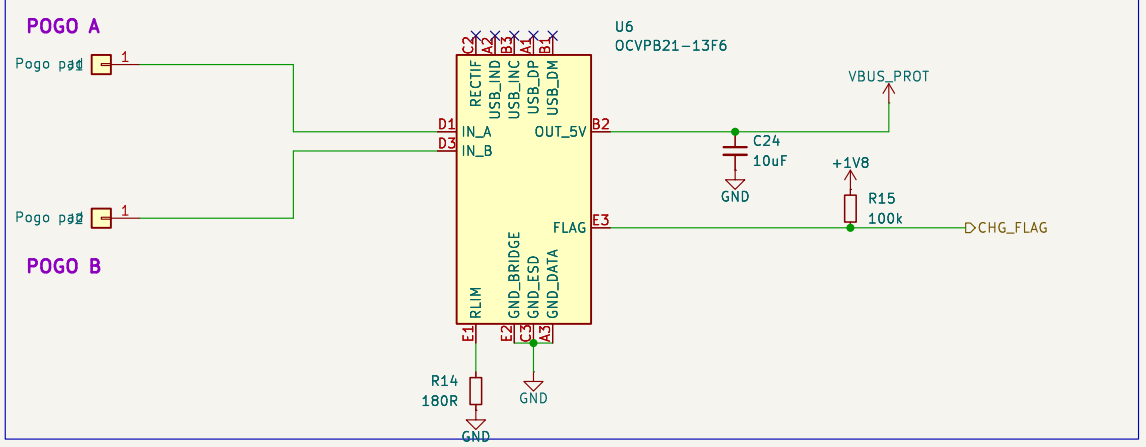
Title: HealthOS Band main board

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 9.0.8		Id: 1/10

CHARGING INPUT — two pogo pads, OCVPB21-13F6, charger

Either polarity is accepted. The protected 5 V output (VBUS_PROT) goes to the charger on the power sheet.

POGO PADS > PROTECTION > VBUS_PROT



OCVPB21-13F6 combines a polarity bridge, ESD clamps, overvoltage cut-off and a current limit. Input range 4.25 to 5.55 V. OUT_5V needs 10 uF or more (C24).

RLIM (R14) sets the current limit: ILIM = (318 − RLIM) / 0.447 mA. 180 Ohm gives about 310 mA.

FLAG is open drain. R15 pulls it up to 1.8 V, so the processor reads a low level when a fault is reported (CHG_FLAG).

Operating temperature of this part is −30 to +65 degC. Keep that in mind for the charging case. The data pins are not used with two pogo pads, so they are left open (crosses). C24 is 10 uF 10 V X5R: its effective value at 5 V is lower, so pick a part that still gives 10 uF or more at 5 V (for example 22 uF).

Done by Harsh Thakur

Sheet: /input_protection/
File: input_protection.kicad_sch

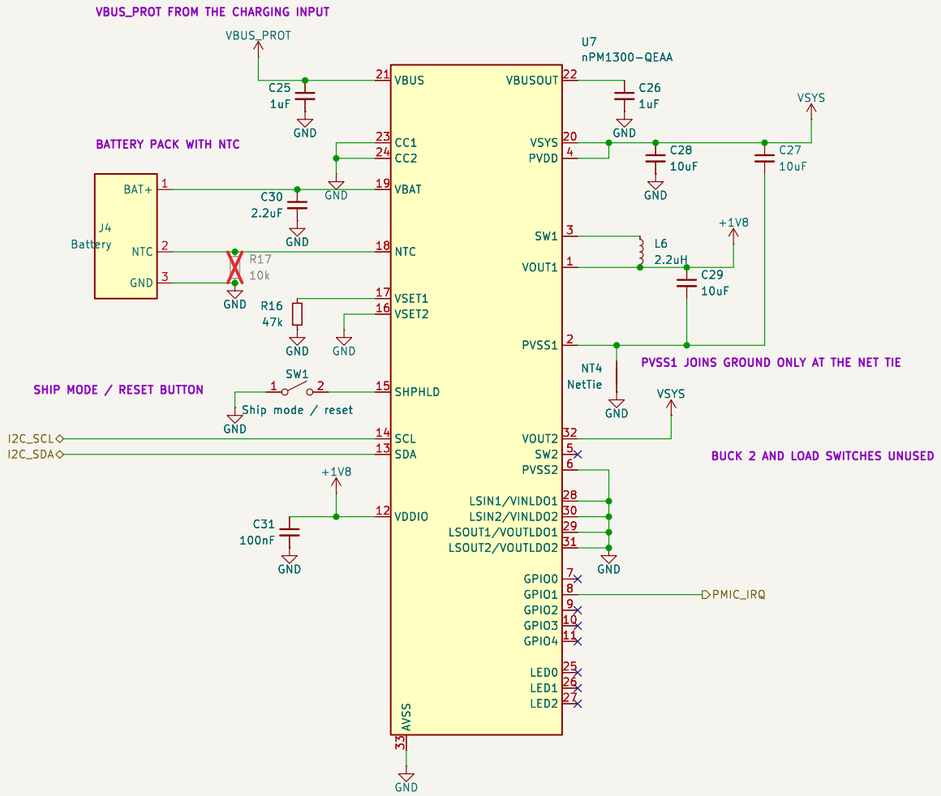
Title: Charging input

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 9.0.8		Id: 2/10

POWER — nPM1300 charger, 1.8 V buck, battery

Charger input VBUS_PROT and the battery come in on the left; VSYS and the 1.8 V rail leave on the right. Nordic reference circuit, configuration 2 (one buck, no USB-C, no LEDs).

nPM1300 — CHARGER, BUCK 1, SYSTEM RAIL



Input: VBUS comes from the OCVPB21 output (VBUS_PROT) with 1 uF (C25). CC1 and CC2 go to ground because there is no USB-C connector. The input current limit starts at 100 mA. The only settings above that are 500 mA and more, but the OCVPB21 limits at about 310 mA, so keep the nPM1300 at its 100 mA default (or set R14 to 51 Ohm for about 600 mA).

Buck 1 = 1.8 V from R16 (47 kOhm on VSET1). L6 2.2 uH (DCR 400 mOhm max, Isat above 350 mA). C29 10 uF (at least 4 uF effective). 200 mA maximum. PVSS1 is the power ground of buck 1; it joins ground only through net tie NT4.

VSYS and PVDD are one net with 10 uF to PVSS1 (C27) and 10 uF to ground (C28). Never feed VSYS from outside. Never power the application from VBAT: use VSYS or VOUT1 only (datasheet).

Battery: the pack must carry its own protection (over/under voltage, over current). A pack with an NTC goes on J4 pin 2. For a pack without NTC fit R17 (10 kOhm) and disable the NTC function in register BCHGDISABLESET. Thermistor type is chosen in software (10, 47 or 100 kOhm).

SW1 on SHPHLD exits Ship mode (a charger on VBUS also does) and a 10 s hold resets the system. TWI address is 0x6B. VDDIO is the 1.8 V rail and has its own 100 nF (C31). GPIO1 is the interrupt line PMIC_IRQ; the other GPIOs, LED0..2 and SW2 are open. The I2C pull-ups are on the sensors sheet.

DECISION NEEDED (rail voltage): the nRF54L15 needs VDD of at least 1.75 V during power-on reset, but 47 kOhm on VSET1 gives 1.8 V +/-5 %, which is 1.71 V at worst. MAX86141 allows 1.7 to 2.0 V. A 1.9 V rail (1.805 to 1.995 V) fits both. Options: keep R16 and have firmware set BUCK1NORMVOUT to 1.9 V at boot, or fit 68 kOhm (2.0 V default) and lower to 1.9 V in firmware. Net name stays +1V8.

Open items: battery cell and connector pin order (J4), button part (SW1), QFN32 exposed pad size (footprint EP 3.45 mm, datasheet 3.5 mm nominal).

Done by Harsh Thakur

Sheet: /power/
File: power.kicad_sch

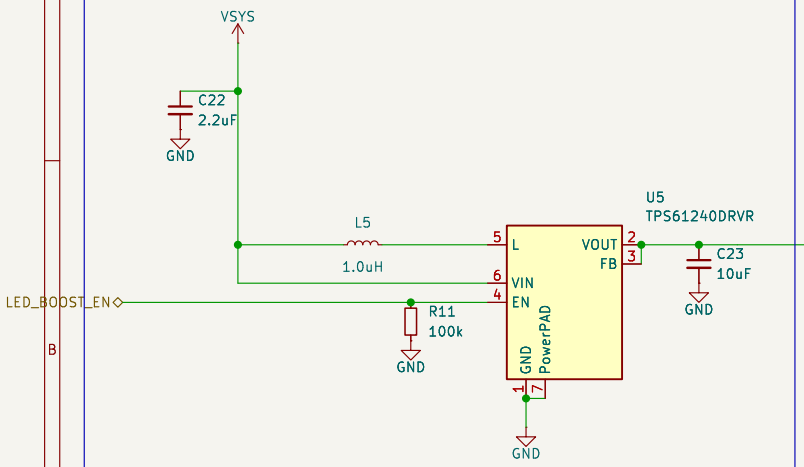
Title: Power nPM1300

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 9.0.8		Id: 3/10

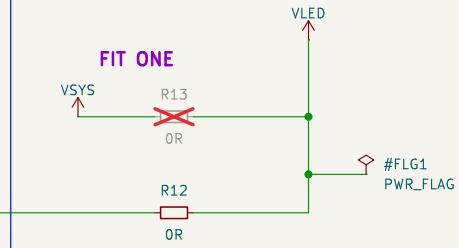
LED SUPPLY — 5.0 V boost (TPS61240) and the VLED link

Battery system rail VSYS goes in on the left; the 5 V boost output or VSYS itself feeds the green LED rail VLED.

TPS61240 — 5.0 V BOOST



VLED — PICK ONE SOURCE



TPS61240 is a fixed 5.0 V boost. The inductor must be 1.0 uH or more (L5, saturation current above 0.9 A). CIN (C22) is 2.2 uF. COUT (C23) must keep at least 2.7 uF effective at 5 V, so a 10 uF 10 V part is used. Keep CIN and COUT close to the pins.

EN needs only 1.0 V, so the 1.8 V logic level from the processor is enough. R11 holds EN low while the processor pins float after reset (nRF54L15 errata 103).

VLED has two possible sources. R12 (boost, fitted) is the default. R13 (VSYS direct, not fitted) is power tree option A for comparison. Fit only one of the two. The PWR_FLAG tells ERC that VLED is driven.

Option A check: VLED must stay inside 3.1 to 5.5 V. The green LED needs its forward voltage (2.8 V max at 20 mA) plus about 0.9 V driver headroom, so about 3.7 V: with a 3.0 to 4.2 V cell the green channel drops out at the bottom of the discharge. Red and IR work down to about 3.1 V. The boost (R12) has no such limit.

Done by Harsh Thakur

Sheet: /led_supply/

File: led_supply.kicad_sch

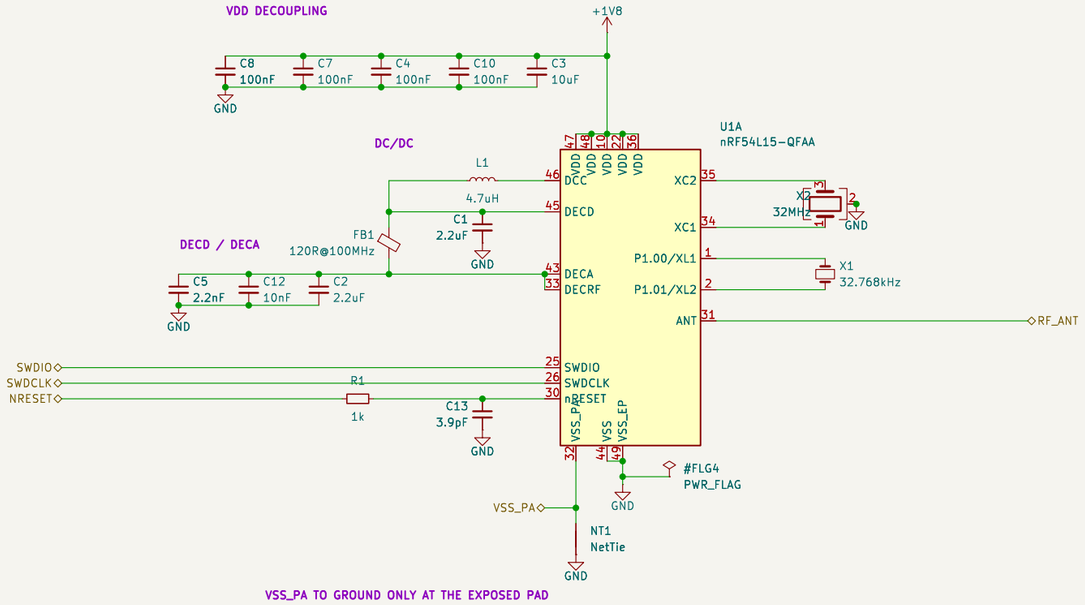
Title: LED supply

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 9.0.8		Id: 4/10

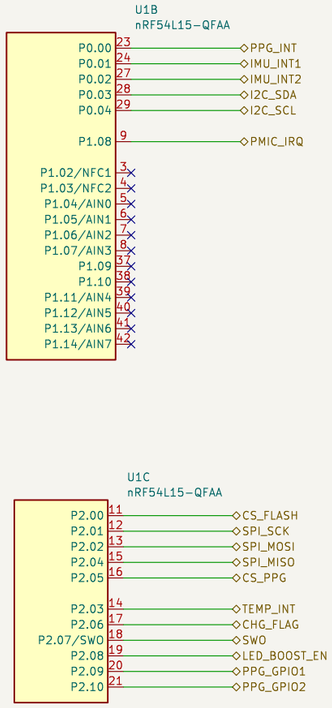
PROCESSOR — nRF54L15 (QFN48): supply, regulators, clocks, reset, debug, GPIO

Nordic circuit configuration 1 (DC/DC from the 1.8 V rail, no NFC), datasheet v1.0 section 10.3 and Table 87. Peripherals sit on the right, the supply network on the left.

nRF54L15 — CORE



GPIO PORTS



Supply: all five VDD pins run from the 1.8 V rail. 10 uF plus four 100 nF as in Table 87. Place each 100 nF next to its VDD pin and keep the 10 uF close by.

Regulators: L1 4.7 uH from DCC to DECD, FB1 from DECD to DECA, C1 2.2 uF on DECD, C2 2.2 uF + C12 10 nF + C5 2.2 nF on DECA (DECRF is tied to DECA).

VSS_PA (pin 32) goes to ground ONLY at the exposed pad, through net tie NT1. The RF network C6 returns to VSS_PA on the top layer (see the RF sheet).

Clocks: 32 MHz crystal on XC1/XC2, 32.768 kHz crystal on P1.00/P1.01. No external load capacitors: internal load capacitors are used (CL 8 pF and 9 pF).

Reset: R1 1 kOhm in series, C13 3.9 pF to ground. SWDIO, SWDCLK and nRESET go to the Tag-Connect pads on the test sheet.

GPIO plan: SPI on P2 (SCK on clock pin P2.01), I2C on P0 (SCL on clock pin P0.04), sensor interrupts on P0. P1.04 to P1.07 and P1.09 to P1.14 are left open because of errata 100. P1.02/P1.03 are NFC and unused. Errata 103: GPIO state is undefined for 10 us at reset, so LED_BOOST_EN has a pull-down.

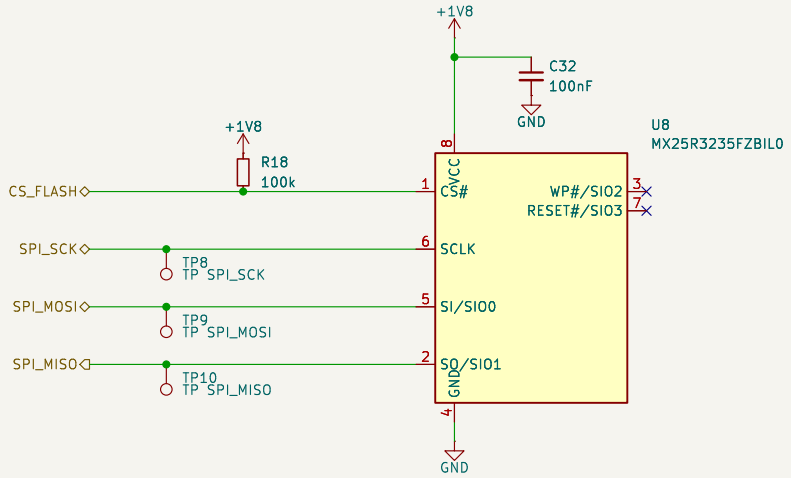
Done by Harsh Thakur

Sheet: /mcu/		
File: mcu.kicad_sch		
Title: Processor nRF54L15		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 9.0.8		Id: 5/10

MEMORY — MX25R3235F 32 Mbit SPI flash

The flash shares the SPI bus with the optical front end. Its chip select CS_FLASH comes from the processor. 1.8 V supply (allowed 1.65 to 3.6 V).

MX25R3235FZBIL0 — 8-LAND USON 4 x 3 mm



Part: MX25R3235FZBIL0, 8-land USON 4 x 3 mm, ultra-low-power default mode, pin 7 = RESET#. VCC 1.65 to 3.6 V, so the 1.8 V rail is fine. 100 nF (C32) close to VCC, as the datasheet recommends.

SPI: mode 0 or 3. Up to 33 MHz for READ and 8 MHz for other commands in ultra-low-power mode, but the MAX86141 on the same bus allows only 4 MHz, so run the bus at 4 MHz or less while the MAX86141 is selected. SO is high impedance while CS# is high, so the bus can be shared (the MAX86141 SDO is also high impedance when CSB is high).

R18 pulls CS# to 1.8 V so the flash stays deselected while the processor pins float after reset (nRF54L15 errata 103) and during power-up and power-down (datasheet section 12).

TP8 to TP10 are test points on the SPI clock, MOSI and MISO lines for the production fixture. WP# (pin 3) and RESET# (pin 7) are left open: both have an internal pull-up while not connected (datasheet section 3, note 1). Use the Reset command if a hardware reset is ever needed.

The two metal pads under the package must stay floating or go to ground only, with no vias or traces under them (datasheet section 19-4). Deep power-down is 0.35 uA maximum.

Open item: land pattern 0.35 x 0.9 mm per pad is drawn from the package outline. Compare with the Macronix recommended land pattern before layout.

Done by Harsh Thakur

Sheet: /memory/
File: memory.kicad_sch

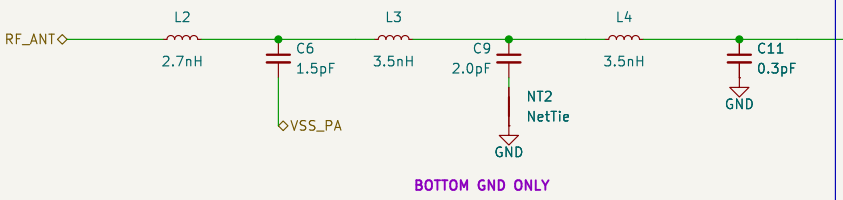
Title: Flash memory

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 9.0.8		Id: 6/10

RF — Nordic match network, antenna tee, chip antenna

The signal runs left to right: ANT pin of the processor, series–shunt–series filter, tee slots, antenna.

NORDIC FILTER AND MATCH



ANTENNA TEE AND ANTENNA



Nordic network: C6 returns to VSS_PA on the TOP layer only. C9 returns to the BOTTOM ground layer only (net tie NT2). Place the whole network as close to the ANT pin as possible. (nRF54L15 datasheet, Fig 180 and Table 87)

Antenna tee: 0R, empty, 0R as a start (R2, C14, R3). The antenna is the Johanson 2450AT18A0100001E in a corner with its keep-out. Tune on the worn band with the final housing. Pin 2 is only a solder anchor. Johanson's own evaluation board uses series 1.0 pF, shunt 2.7 nH, series 3.9 nH: the shunt slot may need an inductor, so the 0402 slot C14 is a generic shunt element.

Done by Harsh Thakur

Sheet: /rf/
File: rf.kicad_sch

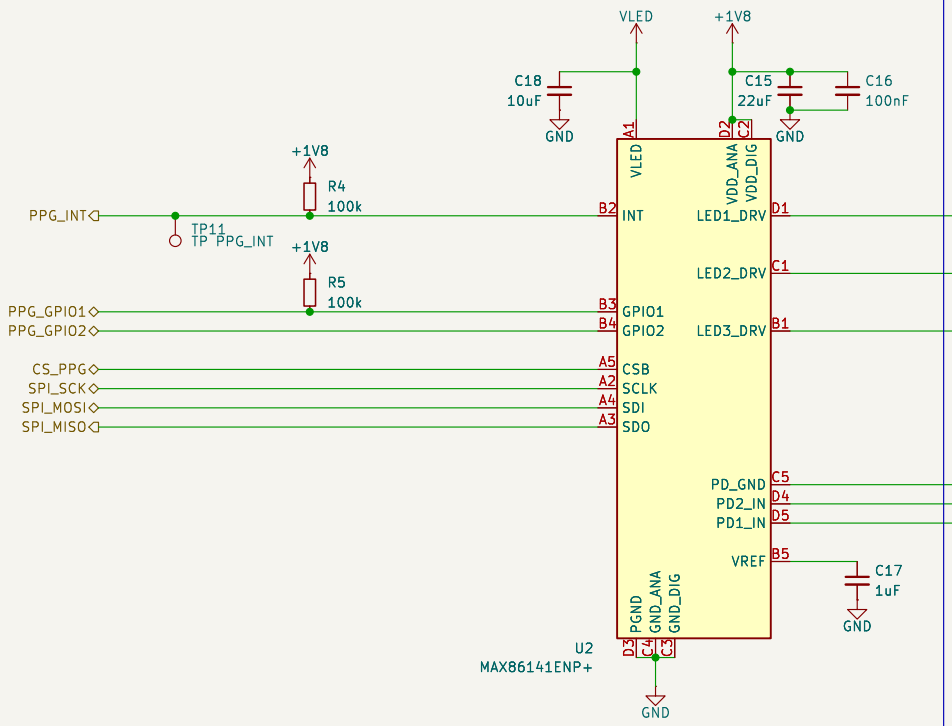
Title: RF match and antenna

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 9.0.8		Id: 7/10

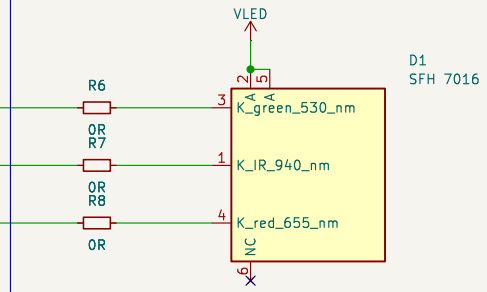
OPTICAL SENSING — MAX86141, SFH 7016 emitter, 2 x VEMD8080

The processor talks to the front end over SPI. The front end drives the three LEDs and reads two photodiodes. Supply 1.8 V, LED supply VLED.

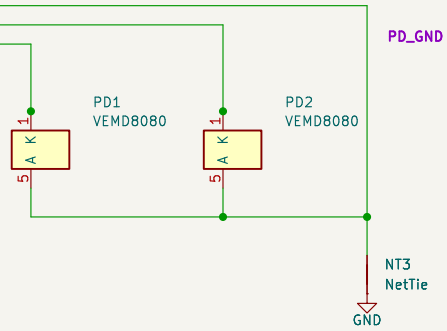
MAX86141 — OPTICAL FRONT END



SFH 7016 — EMITTER



VEMD8080 — PHOTODIODES



Supply: VDD_ANA and VDD_DIG are tied and run from the 1.8 V rail (allowed 1.7 to 2.0 V). 22 uF plus 100 nF on the supply, 1 uF on VREF, 10 uF on VLED. Keep C16 at the VDD_ANA ball. (MAX86141 datasheet p.12 and p.49)

PGND, GND_ANA and GND_DIG meet at one ground. PD_GND is a separate net that touches ground at ONE point only, through net tie NT3 (datasheet layout guidelines p.49).

LED channel order follows MAXREFDES103: LED1 = green, LED2 = IR, LED3 = red. R6, R7 and R8 are 0 Ohm links so the LED current can be measured during bring-up.

PPG_INT and PPG_GPIO1 are open-drain outputs, so R4 and R5 (100 kOhm) pull them up to 1.8 V. SDO is high impedance while CSB is high, so the SPI bus can be shared.

VEMD8080: anode is pin 5, cathodes are pins 1–4 and 6–8 (one net, drawn as one pin). Numbering and pad pattern checked against the Vishay package drawing (top view).

LED limits (SFH 7016): continuous current green 30 mA, red 40 mA, IR 60 mA; pulses 0.75 A, 0.3 A, 1 A at 0.5 % duty. Forward voltage at 20 mA: green 2.2 to 2.8 V, red 1.7 to 2.2 V, IR 1.1 to 1.5 V. Firmware must stay inside these.

SPI: the MAX86141 allows SCLK up to 4 MHz, so keep the bus at or below 4 MHz while it is selected. SDO goes high impedance when CSB is high (shared bus with the flash).

VREF: the datasheet asks for 1 uF (C17). Maxim's MAXREFDES103 board fits 10 uF there. If the noise floor is higher than expected during bring-up, try 10 uF.

Done by Harsh Thakur

Sheet: /optical/
File: optical.kicad_sch

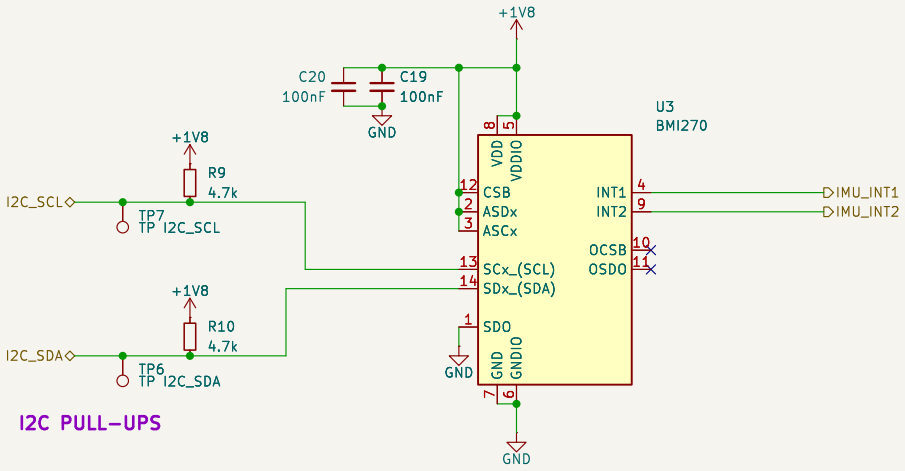
Title: Optical sensing

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 9.0.8		Id: 8/10

MOTION AND SKIN TEMPERATURE — BMI270 + MAX30208 on the I2C bus

Both parts run from the 1.8 V rail and share SDA / SCL. Interrupts go to the processor.

BMI270 — MOTION

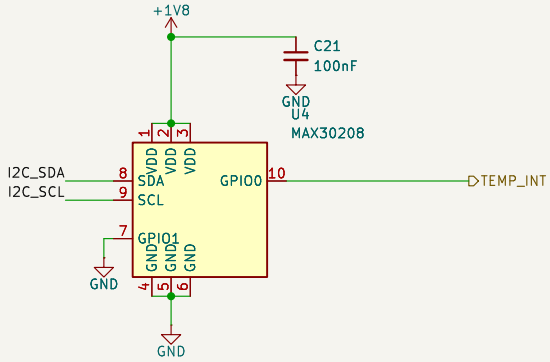


BMI270: I2C address 0x68 because SDO is tied to ground. CSB high selects I2C. ASDx and ASCx (the secondary interface) are tied to VDDIO as the datasheet requires, never to ground. OCSB and OSDO are not used.

C19 and C20 are decoupling capacitors for BMI270 VDD and VDDIO (100 nF each). Keep them close to the pins.

One pair of 4.7 kOhm pull-ups serves the whole bus. At 400 kHz and about 50 pF the rise time is about 0.2 us. The nPM1300 joins this bus on the power sheet.

MAX30208 — SKIN TEMPERATURE



MAX30208: address 0x50. GPIO1 is tied to ground, and GPIO0 reads low at start-up through its internal 1 MOhm pull-down, so the address stays 0x50. GPIO0 is the temperature alarm output (TEMP_INT), open drain and active low in INTB mode: set that mode first, then enable the processor's internal pull-up on P2.03. Do NOT add an external pull-up, it would change the address. All three VDD pins and all three GND pins are connected.

C21 decouples MAX30208 VDD (100 nF). Keep it close to the pins. Read PCN 2204 before ordering.

Done by Harsh Thakur

Sheet: /sensors/

File: sensors.kicad_sch

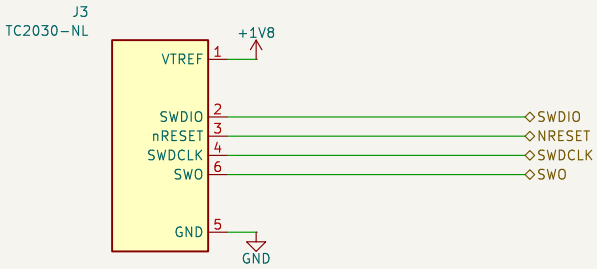
Title: Motion and temperature

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 9.0.8		Id: 9/10

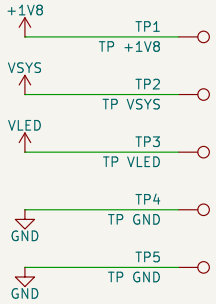
PROGRAMMING AND TEST ACCESS — Tag-Connect SWD pads and test points

The pads are a plug-in cable footprint without a connector. Test points serve the production fixture.

TAG-CONNECT TC2030-NL — SWD



SUPPLY TEST POINTS



Tag-Connect TC2030-NL pin order (to be confirmed against the cable drawing): 1 VTREF, 2 SWDIO, 3 nRESET, 4 SWDCLK, 5 GND, 6 SWO. VTREF takes the 1.8 V rail so the programmer senses the target voltage.

Test points are plain 1.0 mm pads. Put them on one side of the board where the production fixture can reach them (research/storage-nfc-dft/DFT_PROGRAMMING.md). The bus test points sit on the sheets that own the bus: I2C on the sensors sheet, SPI on the memory sheet, PPG_INT on the optical sheet.

Open items: confirm the Tag-Connect pin order, and decide which test points stay on the final board.

Done by Harsh Thakur		
Sheet: /test/ File: test.kicad_sch		
Title: Test and programming		
Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 9.0.8		Id: 10/10